Lecture 3: Standard Form

Week 3: Standard Form

ISE 3200 - Linear Optimization

Why standard form?

- Simplex needs equalities and x >= 0
- Right-hand sides b >= 0

Slack and surplus

- <= constraint: add slack s >= 0
- >= constraint: subtract surplus s >= 0

Free variables and min problems

- x free: x = x+ - x-, both >= 0
- min z = max (-z): negate every coefficient